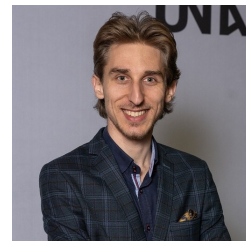


MIHAIL STOIAN

Email: mihail.stoian@utn.de

URL: stoianmihail.github.io



WORK EXPERIENCE

Research Assistant

University of Technology Nuremberg

Data systems lab (Andreas Kipf)

Nov. 2023–present

Nuremberg, Germany

Research Intern

Gray Systems Lab, Microsoft

Robust memory estimation (with Tiemo Bang)

July 2025–Sept. 2025

Barcelona, Spain

Applied Scientist Intern

Amazon Redshift

Learned systems group (Tim Kraska)

July 2023–Oct. 2023

Munich, Germany

Student Research Assistant

TUM, Chair for Database Systems

Umbra: A Flash-Based Database System with In-Memory Performance

Implementing, improving, and testing the functionality

Mar. 2019–Sept. 2023

Munich, Germany

Student Research Assistant

TUM, Chair for Data Analytics and Machine Learning

Graph learning with differential privacy

Jan. 2023–Sept. 2023

Munich, Germany

Research Assistant Intern

Oracle Labs

Graph-in-DB team (Vlad Haprian)

Aug. 2022–Oct. 2022

Zurich, Switzerland

Quantum Software Engineer Intern

Infineon Technologies

Quantum Algorithms group

Mar. 2021–May 2021

Munich, Germany

EDUCATION

University of Technology Nuremberg

PhD, Database Systems

Advisor: Andreas Kipf

Topic: Robust Query Processing

Nov. 2023–present

Nuremberg, Germany

Ludwig Maximilian University

Diploma, Orthodox Theology

Pre-diploma (1.9 / 1.0)

Topics: Early Church History, Liturgics

Oct. 2023–present

Munich, Germany

Technical University of Munich

M.Sc., Elite Software Engineering

Passed with Honors (1.5 / 1.0)

Thesis: *Optimizing Linearized Dynamic Programming*

Supervisor: Thomas Neumann

Oct. 2021–Aug. 2023

Munich, Germany

Technical University of Munich

M.Sc., Informatics

Passed with High Distinction (1.2 / 1.0)

Thesis: *On the Optimal Linear Contraction Order of Tree Tensor Networks, and Beyond*

Oct. 2021–May 2023

Munich, Germany

Supervisor: Christian Mendl

Technical University of Munich

B.Sc., Informatics

Passed with High Distinction (1.2 / 1.0)

Thesis: *An Efficient Implementation of Polynomial-Time Join Ordering*

Supervisor: Thomas Neumann

Oct. 2018-July 2021

Munich, Germany

AWARDS

SIGMOD Research Highlight	2026
<i>DPconv: Super-Polynomially Faster Join Ordering</i>	
SIGMOD Honorable Mention (Best Paper Runner-Up)	2025
<i>DPconv: Super-Polynomially Faster Join Ordering</i>	
EDBT Best Demonstration Award	2025
<i>Virtual: Compresing Data Lake Files</i>	
BTW Best Paper Award	2025
<i>Optimizing Linearized Join Enumeration by Adapting to the Query Structure</i>	
SIGMOD Student Travel Award	2023
<i>Proposal: "Bridging the Gap Between Computational Fields"</i>	
Bronze Medal	2014
<i>National Mathematics Olympiad, Romania</i>	

SCHOLARSHIPS

Deutschlandstipendium	Apr. 2022-Mar. 2023
<i>Allianz SE</i>	
Scholarship awarded by the Ludwig Maximilian University of Munich	
<i>Munich, Germany</i>	

RESEARCH PROJECTS

NVIDIA Research	July 2022-Sept. 2023
<i>Student Research Project</i>	
Einsum optimization on GPU	
Advisors: Jean Kossaifi, Anima Anandkumar	
<i>Remote</i>	
TUM, Visual Computing & Artificial Intelligence Lab	Apr. 2022-Aug. 2022
<i>Practical Course</i>	
Outcome: Twofold improvement over DCP, the deep learning approach for iterative closest point (ICP)	
Advisor: Matthias Niessner	
<i>Munich, Germany</i>	

INTERDISCIPLINARY PROJECTS

INSIGHT	Mar. 2022-Oct. 2022
<i>Chair of Functional Materials (Prof. Peter Müller-Buschbaum)</i>	
Improved INSIGHT performance by 10x. This is the package used by the chair for X-ray measurements	
Published in Journal of Applied Crystallography .	
<i>Munich, Germany</i>	
PushQuantum	Apr. 2021-Aug. 2021
<i>IQM Quantum Computers</i>	
Orbital-Q : Quantum simulations for OLED properties (pitch)	
<i>Munich, Germany</i>	

PROGRAMMING COMPETITIONS

SIGMOD Programming Contest	Feb. 2022-Apr. 2022
<i>ACM SIGMOD</i>	
We implemented a blocking system for Entity Resolution	
Ranking : 6th place, Team: HyTUM	
<i>Munich, Germany</i>	

TECHNICAL SKILLS

Languages: {
 "expert" : { C/C++, Python, SQL, Assembler }
 "advanced" : { Java, Isabelle, HTML/CSS/JS }
}
Frameworks: PyTorch, Spark

LANGUAGE SKILLS

Romanian: Native
English, German: C2
French: C1
Spanish, Ancient Greek: B1

INVITED TALKS

The Case for Cardinality Lower Bounds University of Jena, March 2026
Parachute: Single-Pass Bi-Directional Information Passing. TUMuchData @TUM, January 2026
Instance-Optimized String Fingerprints. Joint Research Seminar, October 2025 (remote)
DPconv: Super-Polynomially Faster Join Ordering. Gray Systems Lab, Microsoft, January 2025 (remote)
Virtual: Compressing World's Parquet Files. TUMuchData @TUM, January 2025
What Selinger Forgot to Tell You About Query Optimization. Joint Research Seminar, June 2024 (remote)
What do databases and tensor networks have in common? University of Jena, August 2023

PREPRINTS

[xBound: Join Size Lower Bounds](#)
Mihail Stoian, Tiemo Bang, Hangdong Zhao, Jesús Camacho-Rodríguez, Yuanyuan Tian, Andreas Kipf
tba
[MLSkip: Data Skipping for ML Filters via Lightweight Metadata](#)
Mihail Stoian*, Mark Gerarts*, Pascal Ginter, Andreas Zimmerer, Jan Van den Bussche, Andreas Kipf
tba

PUBLICATIONS

[Redbench: Workload Synthesis From Cloud Traces](#)
Johannes Wehrstein, Roman Heinrich, **Mihail Stoian**, Skander Krid, Martin Stemmer, Andreas Kipf, Carsten Binnig, Muhammad El-Hindi
VLDB 2026: Experiment, Analysis & Benchmark (EA&B)
[Mind the Gap. Doubling Constant Parametrization of Weighted Problems: TSP, Max-Cut, and More](#)
Mihail Stoian
STACS 2026
[Waiting to Decompress: The Economics of LLM-Based Compression](#)
Andreas Kipf, Tobias Schmidt, Ping-Lin Kuo, Skander Krid, Moritz Rengert, Luca Heller, Andreas Zimmerer, **Mihail Stoian**, Varun Pandey, Alexander van Renen
CIDR 2026
[Parachute: Single-Pass Bi-Directional Information Passing](#)
Mihail Stoian, Andreas Zimmerer, Skander Krid, Amadou Latyr Ngom, Jialin Ding, Tim Kraska, Andreas Kipf
VLDB 2025
[Instance-Optimized String Fingerprints](#)
Mihail Stoian*, Johannes Thürauf*, Andreas Zimmerer, Alexander van Renen, Andreas Kipf
AIDB @VLDB 2025

[Redbench: A Benchmark Reflecting Real Workloads](#)

Skander Krid, **Mihail Stoian**, Andreas Kipf

aiDM @SIGMOD 2025

[DPconv: Super-Polynomially Faster Join Ordering](#)

Mihail Stoian, Andreas Kipf

SIGMOD 2025

[Virtual: Compressing Data Lake Files](#)

Mihail Stoian, Alexander van Renen, Jan Kobiolka, Ping-Lin Kuo, Andreas Zimmerer, Josif Grabocka, Andreas Kipf

EDBT 2025

[Optimizing Linearized Join Enumeration by Adapting to the Query Structure](#)

Altan Birler, **Mihail Stoian**, Thomas Neumann

BTW 2025

[Lightweight Correlation-Aware Table Compression](#)

Mihail Stoian, Alexander van Renen, Jan Kobiolka, Ping-Lin Kuo, Josif Grabocka, Andreas Kipf

3rd Table Representation Learning Workshop (TRL@NeurIPS), 2024

[Unified Mechanism-Specific Amplification by Subsampling and Group Privacy Amplification](#)

Jan Schuchardt, **Mihail Stoian***, Arthur Kosmala*, Stephan Günnemann

37th Conference on Neural Information Processing Systems (NeurIPS), 2024

[On the Optimal Contraction Order of Tree Tensor Networks, and Beyond](#)

Mihail Stoian, Richard Milbradt, Christian B. Mendl

SIAM Journal on Scientific Computing, 2024

[Approximate Min-Sum Subset Convolution](#)

Mihail Stoian

22nd International Workshop on Approximation and Online Algorithms (WAOA), 2024

[DataLoom: Simplifying Data Loading with LLMs](#)

Alexander van Renen, **Mihail Stoian**, Andreas Kipf

Proceedings of the VLDB Endowment, Vol. 17, 2024

[Corra: Correlation-Aware Column Compression](#)

Hanwen Liu, **Mihail Stoian**, Alexander van Renen, Andreas Kipf

2nd Workshop on Cloud Databases (CloudDB @VLDB), 2024

[Fast Joint Shapley Values](#)

Mihail Stoian

Student Research Competition, Companion of the International Conference on Management of Data, 2023

[Faster FFT-based Wildcard Pattern Matching](#)

Mihail Stoian

Student Research Competition, Companion of the International Conference on Management of Data, 2023

[Concurrent Link-Cut Trees](#)

Mihail Stoian

Student Research Competition, Proceedings of the International Conference on Management of Data, 2022

[PLEX: Towards Practical Learned Indexing](#)

Mihail Stoian, Andreas Kipf, Ryan Marcus, Tim Kraska

3rd International Workshop on Applied AI for Database Systems and Applications (AIDB), 2021

[Benchmarking Learned Indexes](#)

Andreas Kipf, Ryan Marcus, Alexander van Renen, **Mihail Stoian**, Sanchit Misra, Alfons Kemper, Thomas Neumann, Tim Kraska

Proceedings of the VLDB Endowment, Volume 14, 2021

[RadixSpline: A Single-Pass Learned Index](#)

Andreas Kipf, Ryan Marcus, Alexander van Renen, **Mihail Stoian**, Alfons Kemper, Tim Kraska, Thomas Neumann

3rd International Workshop on Exploiting AI Techniques for Data Management (aiDM), 2020

[SOSD: A Benchmark for Learned Indexes](#)

Andreas Kipf, Ryan Marcus, Alexander van Renen, **Mihail Stoian**, Alfons Kemper, Tim Kraska, Thomas Neumann

NeurIPS Workshop on Machine Learning for Systems, 2019